

Exercise 1: Overfitting in Action – The Baseline Model

🔗 *Let's start with plain old Linear Regression and see why we need regularization in the first place.*

Steps:

1. Load the dataset (e.g., Boston Housing, or any dataset with numeric features).
2. Preprocess the data: handle missing values, scale numerical features, encode categorical ones.
3. Split into training (80%) and test (20%) sets.
4. Train a **Linear Regression model** (no regularization).
5. Evaluate with **MSE** and **R² Score** on both train and test sets.

🤔 Questions to Think About:

- Is your model **too good** on training data but bad on the test set? (Uh-oh, overfitting!)
 - If test error is way higher than train error, what does that tell you?
 - Can we **fix this mess** without just collecting more data? (Hint: Regularization!)
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🏋️ Exercise 2: Ridge Regression – A Gentle Nudge (L2 Regularization)

💡 *"Let's make the model behave and stop it from going wild!"*

Steps:

1. Train a **Ridge Regression** model (`Ridge()`).
2. Try different values of α (0.01, 0.1, 1, 10, 100).
3. Compare the train/test error and watch how Ridge affects coefficients.
4. Plot **coefficient values** against different α values.

🤔 Questions to Ponder:

- What happens to the model's predictions as α increases?
 - Do the coefficients **shrink** or **disappear**?
 - Does Ridge totally **eliminate** any features? (If not, why?)
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🏋️ Exercise 3: Lasso Regression – The Feature Assassin (L1 Regularization)

💡 "Some features are just noise... let's make them disappear!"

Steps:

1. Train a **Lasso Regression** model (`Lasso()`).
2. Try different α values and see what happens to the coefficients.
3. Identify which features **get dropped** when α is high.
4. Plot **number of non-zero coefficients vs α** .

😞 What's Happening Here?

- Did some features **vanish** as α increased?
 - Why does Lasso **set some coefficients to exactly zero** while Ridge doesn't?
 - If you had a dataset with **1000 features but only 10 were useful**, which method would you pick?
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🧑‍🎓 Exercise 4: Elastic Net – The Best of Both Worlds (L1 + L2 Regularization)

💡 "Why choose one when you can have both?"

Steps:

1. Train an **Elastic Net** model (`ElasticNet()`).
2. Tune α and **`l1_ratio`** (try different values).
3. Compare **Elastic Net vs Ridge vs Lasso** performance.
4. Visualize coefficient values at different **`l1_ratio`** settings.

😞 Let's Discuss:

- What happens when `l1_ratio` = 1? (Spoiler: It's basically Lasso.)
 - What if `l1_ratio` = 0? (Hello Ridge!)
 - When would you **prefer Elastic Net** over the other two?
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🎯 Bonus Challenge: Let's Get Fancy – Hyperparameter Tuning

💡 "Why guess the best α when you can let the computer do it for you?"

Task:

- Use **GridSearchCV** to automatically find the best α and **l1_ratio** for Elastic Net.
 - Compare the test scores of Ridge, Lasso, and Elastic Net.
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- Which regularization method worked **best** for this dataset?
 - How did **cross-validation** help in choosing the best parameters?
 - How would you apply regularization in a **real-world ML project**?